SWARNA BHARATHI INSTITUTE OF SCIENCE & TECHNOLOGY

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TECHNOLOGY EMPOWERS THE NATION

SBIT

Chitti Robot

An Autonomous Humanoid Robot with Sensor-Based Intelligence

Presented at SBIT Engineering College Project Expo

Project Overview

Design Concept

Chitti Robot is designed with a distinctive robotic hulk appearance, combining strength and functionality. The humanoid form factor allows for versatile movement and interaction capabilities.

Core Technology

Built using ESP32 microcontroller with Blynk IoT platform integration, enabling remote control via mobile application. The robot incorporates multiple sensors and actuators for autonomous operation.

Key Features

Ultrasonic Sensor

Distance measurement and object detection up to 200cm range

Mobile Control

Remote operation through Blynk app interface

Voice Module

19-second audio capacity with customizable voice messages

WiFi Connectivity

ESP32-based wireless communication and IoT integration

Intelligent Detection System

1

Person Detection

Ultrasonic sensor continuously monitors proximity within 50cm range

2

Automatic Response

Triggers voice module when object detected

3

Welcome Message

Greets visitors with "Welcome to SBIT Engineering College"

Remote Control Capabilities

Complete directional control through mobile application interface

Forward Motion

Move robot forward through V0 pin control

Reverse Motion

Move robot backward through V1 pin control

Left Turn

Rotate left through V2 pin control

Right Turn

Rotate right through V3 pin control

Technical Specifications

Hardware Components

- ESP32 Microcontroller (WiFi/BT enabled)
- HC-SR04 Ultrasonic Sensor
- Motor Driver IC (L298N equivalent)
- 4 DC Motors for movement
- Voice Module (19-second capacity)
- Power Supply: 12V Battery

Software Platform

- Blynk IoT Application (Mobile)
- Arduino IDE Programming
- WiFi Multi Connection Library

Performance Metrics

Range: 0-200cm detection

Response Time: <500ms

Control Range: WiFi coverage area

Audio: 19-second recording capacity

Power: Continuous operation

Software Architecture

01

WiFi Connection

Establishes connection to specified networks using WiFiMulti library

02

Blynk Initialization

Authenticates with Blynk cloud server using unique token

03

Pin Configuration

Sets up motor control pins (L1, L2, L3, L4) and sensor pins

04

Timer Setup

Configures periodic sensor reading and data transmission

05

Event Loop

Processes commands and sensor data continuously

Code Implementation Highlights

Pin Definitions

```
int l1=26, l2=25, l3=33, l4=32;
int v1=23;
#define echoPin 18
#define trigPin 5
```

WiFi Setup

```
wifiMulti.addAP("SSID", "Password");
while (wifiMulti.run()
  != WL_CONNECTED) {
  delay(250);
  Serial.print('.');
}
```

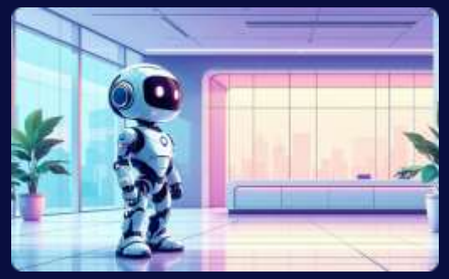
Motor Control

```
BLYNK_WRITE(V0) {
  int pinValue = param.asInt();
  if(pinValue==0) {
    digitalWrite(l1,HIGH);
    digitalWrite(l2,LOW);
    // Forward motion
  }
}
```

Sensor Reading

```
void senddata1() {
 digitalWrite(trigPin, LOW);
 delayMicroseconds(2);
 digitalWrite(trigPin, HIGH);
 delayMicroseconds(10);
 duration = pulseIn(echoPin, HIGH);
 distance = duration/58.2;
}
```

Applications & Future Enhancements



Security & Surveillance

Patrol designated areas and detect intruders with automated alerts

Visitor Guidance

Guide visitors through campus with automated welcome messages

Educational Tool

Demonstrate IoT, robotics, and programming concepts to students

Future Enhancements

- Computer vision integration with camera module
- Speech recognition for voice commands
- Autonomous navigation using pathfinding algorithms
- Extended battery life with power management

Conclusion

Successful Implementation

Chitti Robot successfully demonstrates IoT integration, sensor-based automation, and remote control capabilities with distinctive robotic hulk appearance

Technical Achievement

Combines multiple technologies including WiFi communication, ultrasonic sensing, motor control, and voice output in cohesive system

Practical Value

Provides foundation for security, guidance, and educational applications with potential for further development

Thank you for visiting our project at SBIT Engineering College Project Expo